Market Returns Around VIX Peaks (2016–2025): Replication & Extension

Replicates Zakamulin 2016; analyzes daily S&P 500 returns to test whether VIX spikes predict post-spike rebounds

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Does a VIX Peak Signal a Market Bottom?

Test the “buy the fear” idea using 2016–2025 post-COVID and geopolitical volatility

VIX = **fear index** reflecting expected volatility

Sharp VIX spikes tie to **extreme uncertainty** and financial stress

Common belief: **market bottoms coincide with VIX peaks** → buying opportunity

Study tests this **'buy the fear' hypothesis** across 2016–2025

Focus: post-COVID volatility and **geopolitical risks** shaping unique market environment

Objective: verify whether empirical data support the intuitive buying signal

Research Questions Explored

Assessing VIX peaks, recovery patterns, investor overreaction, and portfolio heterogeneity

Do VIX peaks coincide with abnormally low market returns indicating panic?

Is there a measurable rebound after peaks suggesting short-term recovery?

Do observed patterns reflect investor overreaction creating temporary mispricing?

Do responses vary across portfolio size and sector dimensions?

Methodology & Data Overview

Methodology

- Original Paper Methods: Turning-point algorithms: Lunde-Timmermann filter, Bry–Boschan; Abnormal returns via dummy-variable regressions; Newey–West robust tests for inference
- This Study: Conceptual Replication & extension: Local peak detection using centered rolling maxima; Analyze raw S&P 500 returns + AAR and CAR; Cross-sectional tests across size and sector portfolios; Assess heterogeneous responses to turning points

Data Overview

- Data Sources: VIX and S&P 500 index values (Yahoo Finance); S&P 500 component lists (Wikipedia + GitHub)
- Sample Period: January 2016 – November 2025; Captures modern volatility drivers (COVID-19, inflation, geopolitical events)
- Data Frequency & Rationale: Daily observations → accurate short-term reaction detection; Suitable for peak-based event study analysis

Identifying VIX Peaks: Criteria & Examples

21-day rolling maxima where VIX > **20** — isolates major fear-driven stress episodes (≈15 peaks)

Date	VIX Peak (~)	Event
2020-03-16	85	COVID-19 market shock
2022-06-14	36	Inflation-driven volatility spike
2023-03-10	35	Banking turmoil
2022-02-24	30	Geopolitical tensions
2020-03-09	82	COVID-19 initial crash
2020-03-12	82	COVID-19 volatility surge
2020-03-18	73	COVID-19 continued sell-off
2022-09-21	27	Inflation/market repricing
2023-05-05	23	Regional bank concerns
2021-01-06	37	Political/geopolitical stress
2018-02-05	50	Volatility spike (2018)
2015-08-24	53	Global risk reprice

Event Study: Returns Around VIX Peaks

Quantify S&P 500 short-term reactions within a –10 to +10 trading-day window

Identify Peaks

Locate VIX peaks to center each event window

Define Event Window

Analyze returns from **–10** to **+10** trading days around each peak

Compute AAR

Calculate daily **Average Abnormal Return**: mean S&P 500 return across peaks for each day

Compute CAR

Aggregate returns over the event period to get **Cumulative Abnormal Return** per peak and average across peaks

Aggregate Results

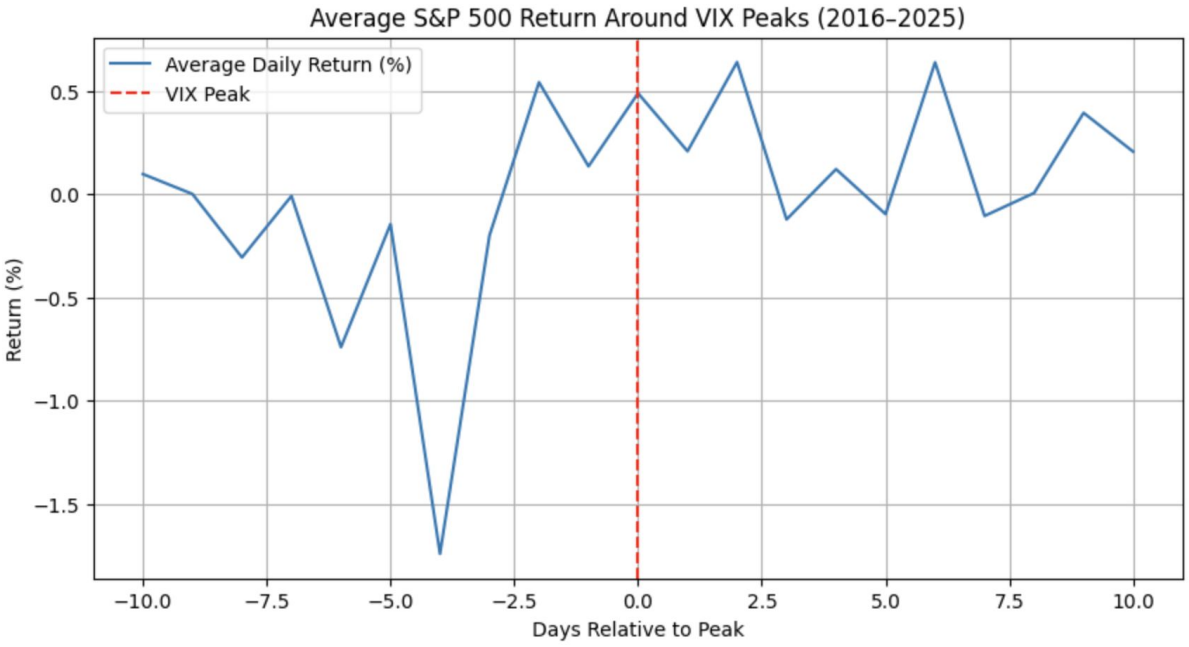
Average AAR and CAR across all peaks to reveal consistent short-term market reactions

AAR Dynamics Around VIX Peak

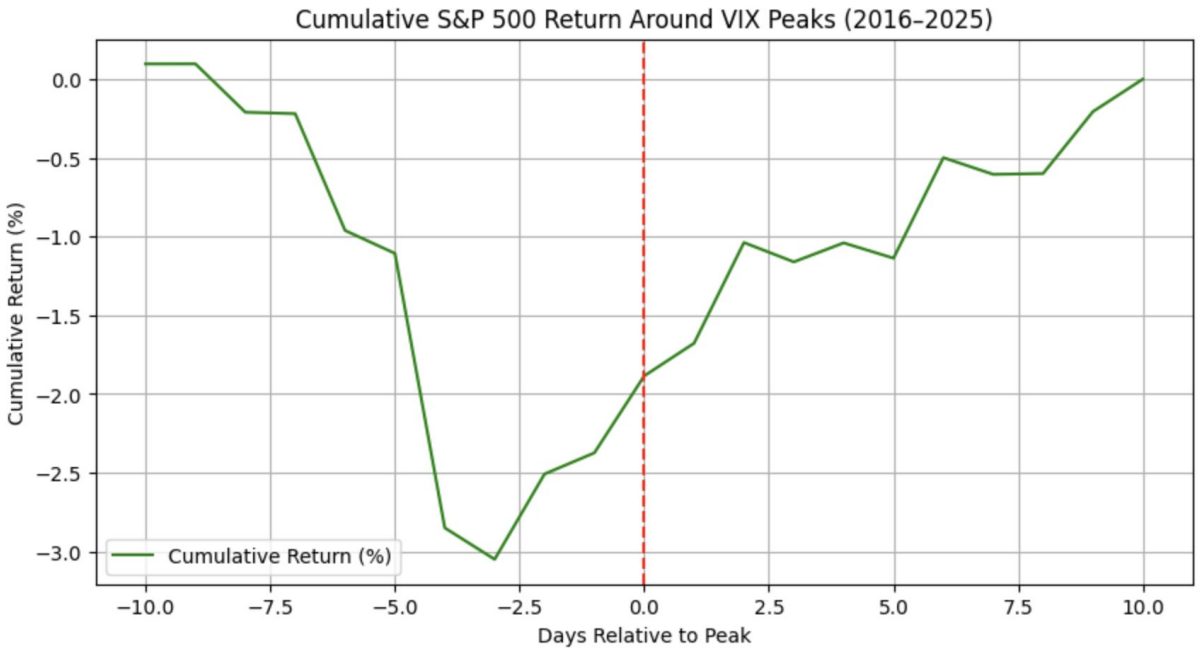
Days -10 to 0: consistent negative AARs, reflecting growing market fear; lowest point at Day 0

Days 1–5: sharp positive rebound in AARs indicating short-term overreaction recovery

Days 5–10: returns gradually stabilize back toward zero Pattern replicates original findings: fear-driven selloffs followed by short-term overreaction



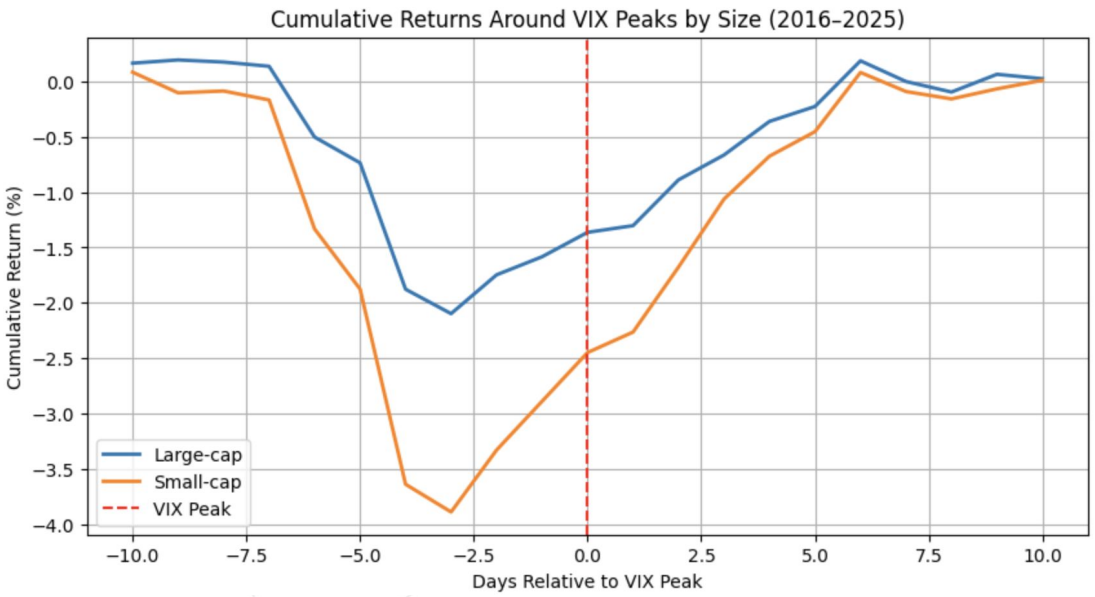
V-shaped CAR pattern around VIX peaks



1. V-shaped pattern in cumulative abnormal returns (CAR) over event window Day -10 to +10
2. CAR falls 3% in the 10 days before the peak — significant selloff
3. CAR recovers 3% after the peak — market correction
4. By Day +10 cumulative returns settle near zero — partial, incomplete recovery
5. Pattern aligns with original paper's observation of return reversals after VIX peaks

Small Caps: Bigger Drops, Stronger Rebounds

Cross-sectional CARs around VIX peaks — sentiment-driven overreaction in small caps

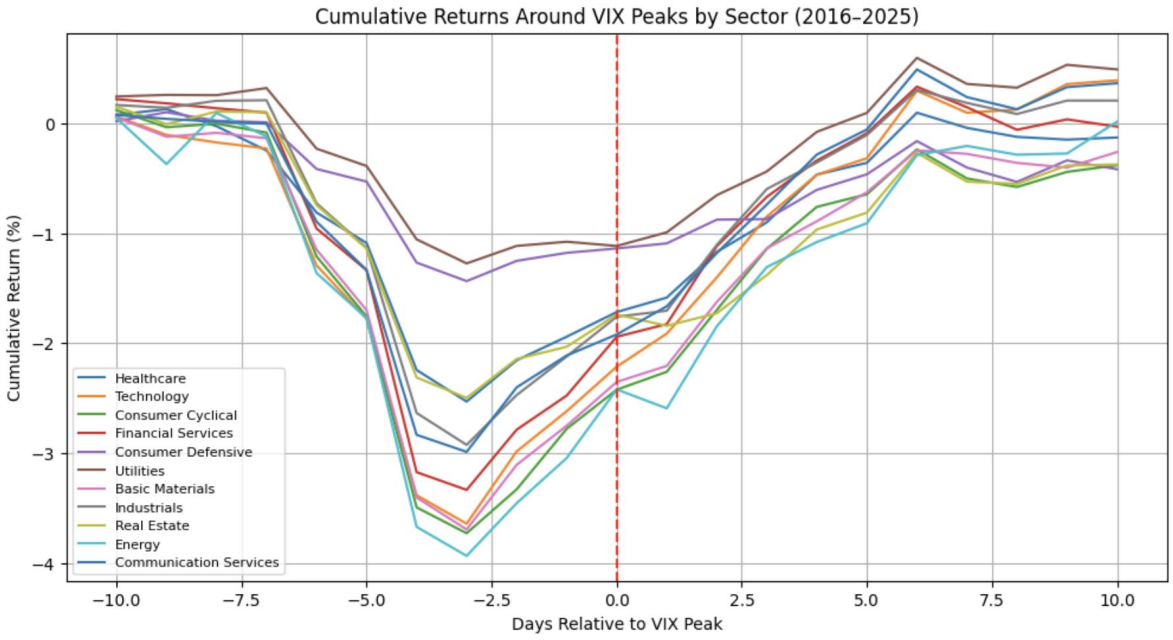


1. **Small caps drop ~3.8%** in cumulative returns before VIX peaks — the deepest decline among size groups
2. **Stronger post-peak rebound** in small caps, showing pronounced recovery once fear subsides
3. **Large caps** show **milder pre-peak drops** and **weaker recoveries**, reflecting lower sensitivity to volatility shocks
4. CAR visual compares **small vs large cap cumulative abnormal returns** around VIX peaks
5. Pattern is **consistent with prior research** linking size and volatility to amplified behavioral effects
6. Implication: **sentiment-sensitive small caps** are more prone to **overreaction** during high-fear periods

Sector CARs: Synchronized Drops, Divergent Recoveries

All sectors fall before VIX spikes; high-beta recover faster, defensive sectors lag

1. All sectors show synchronous drops before VIX peaks — broad market fear
2. Energy and Technology (high-beta) exhibit quicker, stronger recoveries after the drop
3. Utilities and Real Estate (defensive) show slower rebounds and muted correction behavior
4. Sectors with higher market-risk exposure show stronger overreaction and correction dynamics
5. Summary: synchronized stress, followed by divergent recovery speeds — sector beta helps anticipate rebound strength



Methodological Differences vs. Original Paper

Concise comparison of choices and confirmatory outcome

1. **Sample universe:** used **S&P 500** rather than total market coverage
2. **Return metric:** analyzed **raw returns** instead of abnormal returns
3. **Peak identification:** used rolling maxima vs formal peak-dating algorithms
4. **Timeframe:** covered **2016–2025** vs **1990–2015** in original
5. **Cross-section:** substituted **sector portfolios** for value/growth portfolios
6. **Outcome:** core results and behavioral conclusions remain consistent

Key Limitations That Temper Interpretation

Concise constraints that affect generalizability and significance

1. Extraordinary post-2020 market conditions **(COVID-19, interest-rate shocks) may reduce generalizability**
2. S&P 500 sample bias **limits representation of small-cap and growth stocks**
3. **Use of** raw returns **may underestimate abnormal return magnitudes**
4. Peak detection **may include milder spikes not matching formal peak definitions**
5. **No** Newey–West **or similar robust tests applied; significance claims are tempered**

Practical Lessons from VIX Study

Actionable implications for quantitative finance and risk management

VIX spikes = fear-driven selloffs, typically followed by partial rebounds — expect short-term overreactions and corrections

High-beta and small-cap portfolios show stronger return volatility — greater sensitivity to investor sentiment

VIX remains a timely sentiment indicator — useful for short-term abnormality detection and risk signals

Apply VIX signals in **quantitative strategies and risk management**: position sizing, hedging, and stress testing

Treat spikes as **signal, not certainty** — combine VIX with cross-sectional (beta, cap) and market-confirmation metrics

Replication Confirms Behavioral Finance Explanation

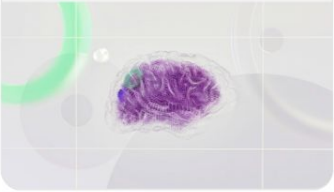
Modern dataset validates Zakamulin's core findings on VIX-driven short-term anomalies

Replication success: modern dataset replicates Zakamulin's core findings

VIX peaks as signal: VIX spikes predict short-term abnormal market behavior

Behavioral patterns: persistent overreaction and partial reversal effects

Model implication: behavioral finance explains dynamics better than purely rational models



Practical value: supports integrating sentiment measures like **VIX** into quant research and asset management

Questions and Discussion

Share questions on methodology, findings, limitations, or implications of the VIX-peaks market-returns study

Name	Contributions
Abhinn Tejpal Sharma	*Code and Presentation*
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